MILITARY SPECIFICATION
TEST SET, MISSILE AUXILIARIES
TS-3479( )/AWM-23(V)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Missile Auxiliaries Test Set (MATS) TS-3479( )/AWM-23(V).

1.2 Classification. The MATS shall consist of the following items:

a. Assembly, Eject Cable
b. Assembly, Test Set

1.3 Associated equipment. The MATS shall operate with the cable assembly set of MIL-C-81954(AS).

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) is the applicable general specification for the MATS described herein (see 6.6).

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4935

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MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)
MIL-C-81954(AS)	Cable Assembly Set, Electrical, MATS, Special Purpose

Naval Air Systems Command (NAVAIR)

AS-2695	Interface Specification, F-14A Weapon Control System AN/AWG-9 (N-3) Interface with the AIM-54A Missile
AS-2696	Interface Specification, AIM-54A Guided Missile (Production), Installation and Physical Interface with F-14A Aircraft

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-761	Electric Power, Alternating Current for Shipboard Use, Characteristics and Utilization of

PUBLICATIONS

Hughes Aircraft Company

3282209	Acceptance Test Procedure, Missile Auxiliary Test Set

2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract, The contract shall have precedence over any other document.
b. This specification. This specification shall have precedence over all applicable subsidiary documents (see 6.6).
c. Referenced documents. Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

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3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6).

3.2 Design and construction. Applicable except as follows.

3.2.1 Design features. Applicable. The MATS shall consist of a combination case in accordance with MIL-T-21200 containing an umbilical connector and wiring harness, control panel and readout, electronic modules, eject cable assembly, cover, and a launcher attachment device. The cover shall provide storage space for the eject cable, and a brief operating instruction plate. The requirements for connections for external power of MIL-T-21200 shall not apply.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Not applicable.

3.2.1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight limitations. Dimensions of the MATS shall not exceed 21 by 11.5 by 10.5 inches. The weight shall not exceed 25 pounds.

3.2.1.9.1 Storage space. Storage space shall be provided for an eject cable assembly in the case cover.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. Secondary power forms shall be provided by power supplies which are a part of the MATS.

3.2.1.12 Interface. The MATS shall directly attach to the AIM-54A launcher umbilical connector and the eject cartridge connector in accordance with AS-2696 (see 6.9). The MATS-shall be capable of being connected to the F-14A aircraft umbilical and attack connectors at the aircraft/weapon rail interface and to the Module Test Station (MTS) OJ-217( )/AWM-23(V) and OJ-378( )/AWM-23(V) by the cable set of MIL-C-81954(AS).

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3.2.2 Utilities. Applicable except as follows.

3.2.2.1 Electrical power. See 3.2.3.1.

3.2.2.2 Clean air. Not applicable.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Not applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. The MATS shall meet the performance requirements of this specification when supplied with 115 volts, 400 Hertz (Hz), 2.0 amperes maximum through the launcher Umbilical connector or the cable set. The MATS shall operate with electrical power having the characteristics of MIL-STD-761 Type I.

3.2.3.2 Environmental. Applicable.

3.2.3.2.1 Organizational level. Applicable.

3.2.3.2.2 Intermediate level. Not applicable.

3.2.4 Maintainability. Applicable except as follows.

3.2.4.1 Mean time to repair. Applicable,

3.2.4.2 Test points. Not applicable.

3.2.4.3 Shop replaceable assembles. Applicable.

3.2.4.4 Self test capabilities. Not applicable.

3.2.5 Reliability. The mean time between failures shall be 500 hours minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

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3.2.6 Electromagnetic interference (EMI). Applicable, except as follows:

a. MIL-STD-461, test CE02 and the limits therein shall apply to the eject cable only.
b. MIL-STD-461, tests CE01, CE03, CE04, CE05, CS01, CS02, CS06, RE03, RS01, RS02, and RS04 shall not apply.

3.2.7 Utilities protection. Applicable except as follows:

3.2.7.1 Liquid cooling protection. Not applicable.

3.2.7.2 Clean air protection. Not applicable.

3.2.7.3 Electrical power protection. Applicable.

3.2.8 Calibration cycle. Applicable.

3.2.9 Power switching control. The MATS shall provide the necessary switching to control incoming power forms. A front panel POWER ON switch shall control incoming power, and may be utilized for emergency shut-down.

3.2.10 Status displays. An illuminated numerical display shall provide an indication of test status.

3.3 Performance.

3.3.1 General. The MATS shall provide the capability to test the tactical interface to the AIM-54A Guided Missile through the F-14A launcher missile umbilical connector. In addition, the MATS shall provide capability to test the eject cartridge firing signal at the eject cartridge connector. The functional interface is defined in AS-2695. When used with the cable set provided in accordance with MIL-C-81954(AS), the MATS shall provide the capability to test the tactical interface of the F-14A aircraft with the AIM-54A missile weapon rail removed. The MATS shall be capable of being functionally tested at the MTS when connected to the MTS by the cable set.

3.3.2 Organizational level functional requirements. Applicable.

3.3.3 Intermediate level functional requirements. Not applicable.

3.3.4 Warmup time. Applicable.

3.4 Detailed functional requirements.

3.4.1 Test capabilities. The following test capabilities shall be provided. Failures shall be indicated on the display. A unique numerical indication shall be provided to identify each individual umbilical line in the event of a malfunction.

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3.4.2 Power verification. Verify the presence of 115V rms, 400 Hz, single phase power from an AN/AWG-9.

3.4.3 Missile parameter signals. Verify the presence/absence of each missile parameter digital signal.

3.4.4 Analog output (AO) from digital computer. Verify the presence of each AO signal.

3.4.5 Digital test reports. Simulate missile digital test report outputs.

3.4.6 Rate integrating gyro (RIG) signals. Verify the presence of each RIG signal including pitch and roll initial conditions.

3.4.7 Identification (ID) and interlock signals. Simulate the missile identification output.

3.4.8 Slaving reference voltage and seekerhead positioning (SHP) signals. Perform the following:

a. Verify the presence of slaving reference voltage.
b. Verify the presence of three seekerhead positioning signals.
c. Test i, j, and k SHP signals for proper phase using the slaving reference voltage as a phase reference.
d. Verify the continuity of associated return lines.

3.4.9 Missile power supply outputs. Verify the following:

a. The presence of the +100 Volt Ready, Launch, MOAT (RLM) input, the +100 Volt launch Missile On Aircraft Test (MOAT) A input, and the +100 Volt MOAT B input.
b. The continuity of +100 Volt RLM return and +100 Volt launch - MOAT return lines.

3.4.10 Analog test reports. Perform the following;

a. Simulate analog test report outputs.
b. Generate simulated missile active frequency identification,

3.4.11 MOAT signals. Verify the presence/absence of:

a. External slaving command
b. Timer zero set
c. Velocity servo enable
d. MOAT seeker test
e. Parameter reset.
f. Eight second signal
g. Center channel target simulator

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h. Pitch and yaw rate test
i. Transmitter rf on/off
j. Roll flipper command
k. Clock pulse
l. Doppler g-bias
m. Verify the receipt of semiactive channel select signals A, B, and C.

3.4.12 Miscellaneous signals. Verify the following:

a. The presence/absence of prelaunch doppler frequency signals
b. The presence/absence of single target track enable signal
c. The presence/absence of ACM active launch signal
d. The presence of 12 Volt ID power
e. The presence of 12 Volt motor ignition hold

3.4.13 Irreversible signals. Verify the presence of the following signals:

a. Battery activate
b. Pull pin unlock
c. Pull pin relock
d. Eject cartridge firing signal. Access to the eject cartridge connector shall be in accordance with AS-2696 (see 6.9).

3.4.14 Frequency coded firing signals. Circuitry shall be provided to verify the frequency coded firing signals at the aircraft/weapon rail.

3.5 Workmanship. Applicable.

3.6 Elements.

3.6.1 Elapsed time meter. An elapsed time meter conforming to MIL-M-7793 shall be provided. The meter shall be readily visible during operation of the MATS and shall be capable of registering at least 9,999 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

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4.2.5 Acceptance test plan. Applicable. Table I shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in-test. Applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

TABLE I

ACCEPTANCE TEST PLAN FOR MISSILE AUXILIARIES TEST SET

TS-3479( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate the functional performance of the Missile Auxiliaries Test Set.

2.0 TEST CONDITIONS

a. Test Equipment

(1) MTS/MATS Drawer Test Position (HAC Part No. T493457-11-1)

b. Utilities Required (Information only, ref 4.2.8)

(1) Power as specified in 3.2.2.1

3.0 TESTS. In accordance with Acceptance Test Procedure 3282209.

a. Perform the acceptance test.
b. Perform operational run-in-tests.

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5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The MATS is intended to be portable and for use at the organizational level to check out the electrical interface to the AIM-54A Guided Missile through the F-14A launcher missile umbilical connector. In addition, the MATS provides the capability to test the eject cartridge firing signal at the eject cartridge connector, When used with the cable set of MIL-C-81954(AS) the MATS will check out the electrical interface to the AIM-54A missile with the missile weapon rail removed. The MATS will be capable of being functionally tested at the intermediate level when connected to the MTS by the cable set.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross reference. The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

	Official Nomenclature	Contractor Part No.
a.	Test Station Module OJ-217( )/ AWM-23(V) and OJ-378( )/AWM-23(V)	493457-1XX
b.	Test Set, Missile Auxiliaries TS-3479( )/AWM-23(V)	493502-1XX
c.	Cable Assembly Set, Electrical, MATS, Special Purpose	493513-1XX

6.5 Liquid cooler. Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-T-81781 (AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification. When the term “not applicable” is used in this specification it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms, Applicable. In addition, the following shall apply:

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ACM	Air combat mode
AO	Analog output
ID	Identification
MATS	Missile auxiliaries test set
MOAT	Missile on aircraft test
MTS	Module test station
RIG	Rate integrating gyro
RLM	Ready, Launch, MOAT
SHP	Seekerhead positioning

6.8 Convenience outlet power. Not applicable.

6.9 F-14A/MATS interface. Dimensional characteristics and details of the mounting and Interconnection interface of the MATS to the F-14A are in accordance with AS-2696. The F-14A contractor will provide suitable physical Interconnection access for the MATS.

6.10 Changes from previous issue are not marked.

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-81781D(AS) Test Set, Missile Auxiliaries TS-3479( )/AWM-23(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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